

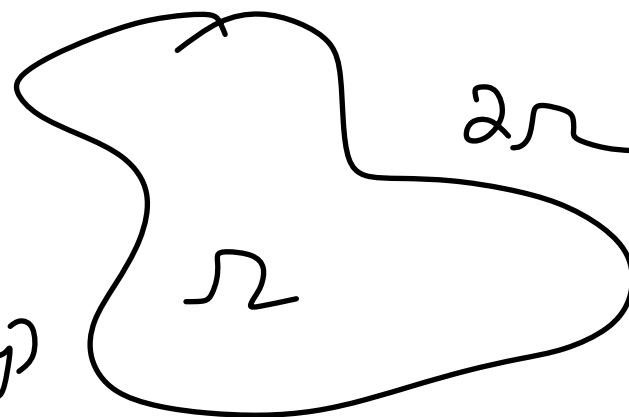
Fast multipole methods

So far

$$\Delta u = 0 \quad \text{in } \Omega$$

$$u = f \quad \text{on } \partial\Omega$$

$$u = D[\sigma](x) = \frac{1}{2\pi} \int \frac{(x-y) \cdot n(y)}{|x-y|^2} \sigma(y) dy$$



$$= \operatorname{Re} \left(\frac{1}{2\pi i} \int_{\partial\Omega} \frac{\sigma(\xi)}{z - \xi} d\xi \right)$$

$$z = x_1 + i x_2$$

$$\xi = y_1 + i y_2$$

Impose boundary conditions

$$-\frac{1}{2}\sigma + D^{\text{PV}}[\sigma] = f$$

Discretize

$$-\frac{1}{2}\sigma_j + \sum_{\substack{k=1 \\ k \neq j}}^N \frac{\sigma_k w_k}{(z_j - z_k)} = f_j$$

$$A \vec{\sigma} = \vec{f}$$

$$A_{jk} = -\frac{1}{2} \delta_{jk} + \frac{w_j}{z_j - z_k}$$

$$A\vec{\sigma} = \vec{f}$$

$$A = -\frac{1}{2} \delta_{jk} + \frac{w_j}{z_j - z_k}$$

How to solve linear system?

Since well-conditioned $\Rightarrow$ can use iterative solver ($\times$)

Requires computing $A\vec{b}$ where $\vec{b}$ is given

Cost to compute solⁿ

Nit $\times$ N^2

Can we do better?

Number of GMRES iterations

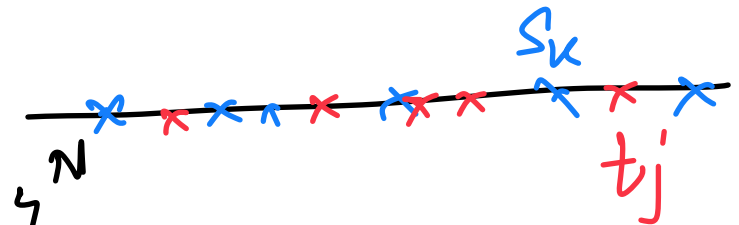
Cost of applying matrix densely

(Typically $O(N)$ due to $\times$)

Dense matrices are not data dense

$$A_{jk} = \cos(t_j - s_k)$$

$\{t_j\}_{j=1}^M, \{s_k\}_{k=1}^N$



A is an $M \times N$ matrix

Naive matrix cost is $O(MN)$

$$f_j = \sum_k A_{jk} \sigma_k = \sum_{k=1}^N \cos(t_j - s_k) \sigma_k$$

$$= \sum_{k=1}^N \cos(t_j) \cos(s_k) \sigma_k + \sin(t_j) \sin(s_k) \sigma_k$$

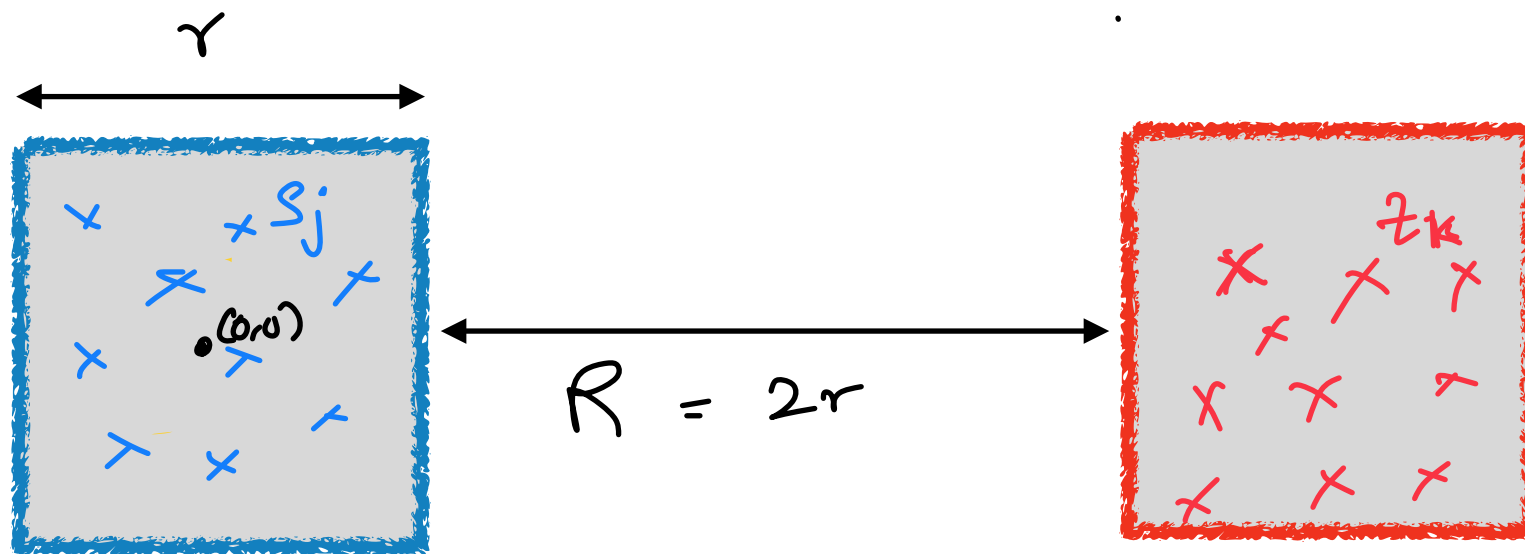
$$= \cos(t_j) \sum_{k=1}^N \cos(s_k) \sigma_k + \sin(t_j) \sum_{k=1}^N \sin(s_k) \sigma_k$$

Step 1: Compute $W_1 = \sum_{k=1}^N \cos(s_k) \sigma_k$ $W_2 = \sum_{k=1}^N \sin(s_k) \sigma_k$
 $O(N)$

Step 2: $f_j = \cos(t_j) W_1 + \sin(t_j) W_2$
 $O(M)$

$$\begin{bmatrix} \square \\ \square \end{bmatrix}_{M \times 2} = \begin{bmatrix} \square \\ \square \end{bmatrix}_{2 \times N}$$

Exact Rank-2 matrix



$$\phi(t_k) = \sum_{j=1}^N \frac{\sigma_j}{t_k - s_j}, \quad k = 1, 2, \dots, M \quad \text{Naive work: } O(NM)$$

$$\phi(t) = \sum_{j=1}^N \frac{\sigma_j}{t - s_j} = \sum_{j=1}^N \frac{\sigma_j}{t(1 - \frac{s_j}{t})}$$

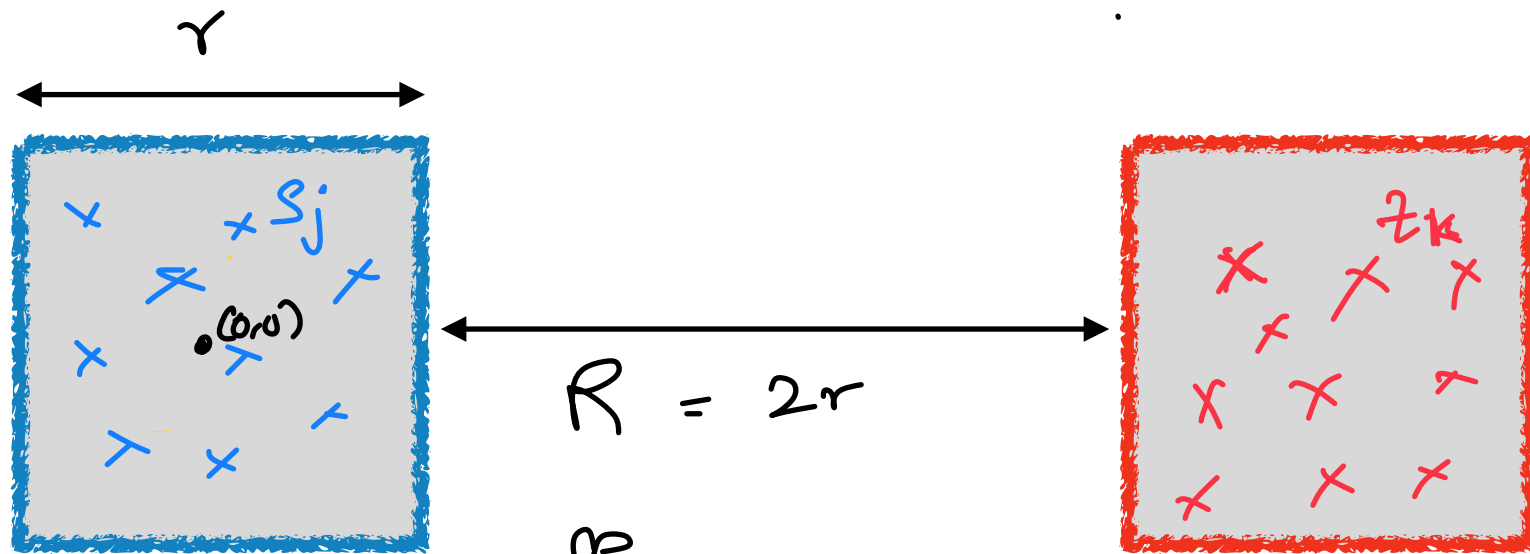
$|t| > r + R$
 $|s_j| \leq \sqrt{2}r$

$$= \sum_{j=1}^N \sum_{l=0}^{\infty} \frac{\sigma_j}{t} \left(\frac{s_j}{t}\right)^l$$

$\frac{1}{1-x} = 1 + x + x^2 + \dots$

$$= \sum_{l=0}^{\infty} \frac{1}{t^{l+1}} \underbrace{\sum_{j=1}^N \sigma_j (s_j)^l}_{\alpha_l}$$

(Multipole expansion of ϕ)



$$\phi(t) = \sum_{j=1}^N \frac{\sigma_j}{t - s_j} = \sum_{l=0}^{\infty} \frac{\alpha_l}{t^{l+1}} \quad \alpha_l = \sum_{j=1}^N \sigma_j (s_j)^l$$

Approximate ϕ by $\tilde{\phi}(t)$

$$\tilde{\phi}(t) = \sum_{l=0}^p \frac{\alpha_l}{t^{l+1}}$$

$|t| \geq R + r = 3r$
 $\frac{1}{|t|} \leq \frac{1}{3r}$
 $|s_j| \leq \sqrt{2}r$

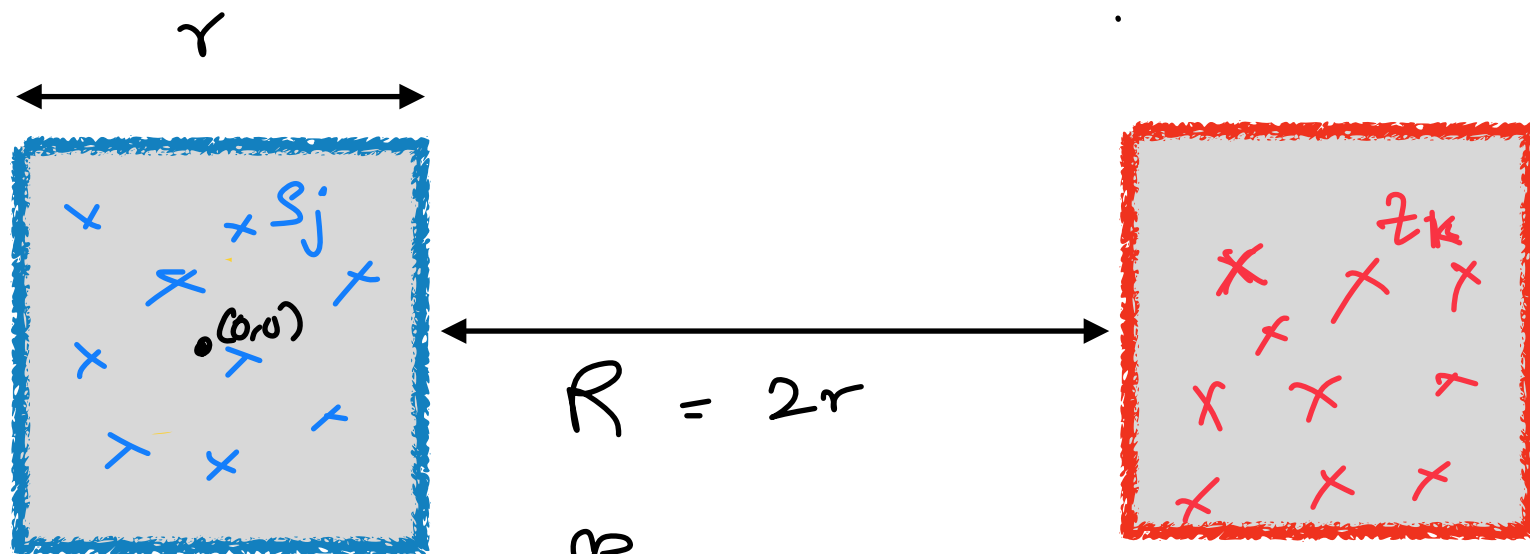
Then $E = |\phi - \tilde{\phi}| = \left| \sum_{l=p+1}^{\infty} \frac{\alpha_l}{t^{l+1}} \right| = \left| \sum_{l=p+1}^{\infty} \sum_{j=1}^N \frac{\sigma_j}{t^{l+1}} \left(\frac{s_j}{t} \right)^l \right|$

$$\leq \frac{\sum_{j=1}^N |\sigma_j|}{3r} \sum_{l=p+1}^{\infty} \left(\frac{|s_j|}{|t|} \right)^l \leq \frac{\sum_{j=1}^N |\sigma_j|}{3r} \sum_{l=p+1}^{\infty} \left(\frac{\sqrt{2}r}{3r} \right)^l$$

$$\leq \frac{\sum |\sigma_j|}{3r} \left(\frac{\sqrt{2}}{3} \right)^{p+1} \left(\frac{1}{1 - \frac{\sqrt{2}}{3}} \right)$$

Error is decaying exponentially in p !

$p = 20, \quad \frac{E}{\sum |\sigma_j|} \leq 10^{-6}$



$$\phi(t) = \sum_{j=1}^N \frac{\sigma_j}{t - s_j} = \sum_{l=0}^{\infty} \frac{\alpha_l}{t^{l+1}}$$

Approximate ϕ by $\tilde{\phi}(t)$

$$\tilde{\phi}(t) = \sum_{l=0}^p \frac{\alpha_l}{t^{l+1}}$$

$$\alpha_l = \sum_{j=1}^N \sigma_j (s_j)^l$$

$$|t| \geq R + r = 3r$$

$$\frac{1}{|t|} \leq \frac{1}{3r}$$

$$|s_j| \leq \sqrt{2}r$$

Cost of computing $\phi(t_k) = O(NM)$

Cost of computing $\tilde{\phi}(t_k)$

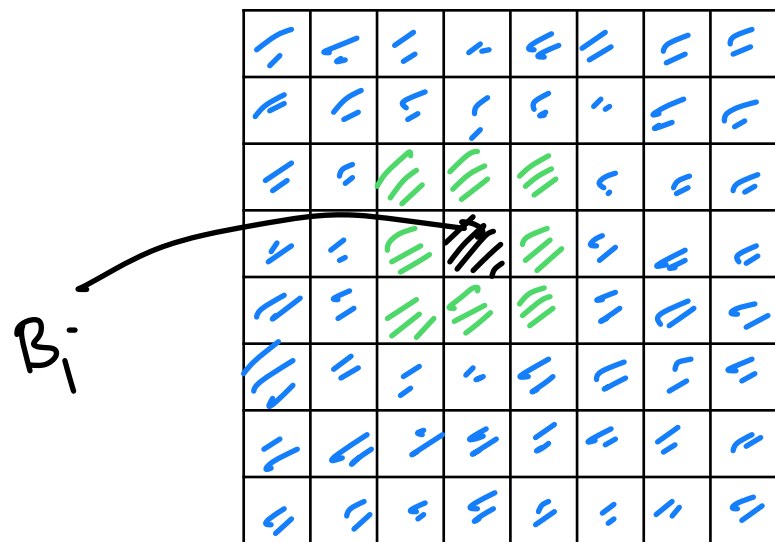
Step 1: Compute

$$\alpha_l = \sum_{j=1}^N \sigma_j (s_j)^l \quad l = 0, \dots, p \quad O(Np)$$

Step 2: Evaluate

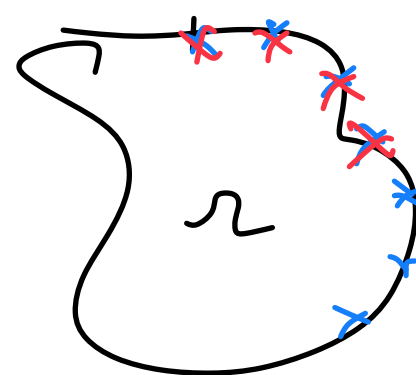
$$\tilde{\phi}(t_k) = \sum_{l=0}^p \frac{\alpha_l}{(t_k)^{l+1}} \quad k = 1, \dots, M \quad O(Mp)$$

$\tilde{\phi}$ a "low-rank approximation" of ϕ in this case!



$$\phi(x_k) = \sum_{\substack{j=1 \\ j \neq k}}^N \frac{\sigma_j}{x_k - x_j}$$

Consider $x_k \in B_i$



$$x_k' = x_j'$$

Points are not well-separated

$$\equiv B_j \in \text{Near}(B_i)$$

$$\equiv B_j \in \text{Far}(B_i)$$

Idea: For all boxes $B_j \in \text{Far}(B_i)$ replace the contribution of $\phi(t)$ in that box by the corresponding multipole expansion (truncated)

$$\phi(x_k) = \sum_{B_l} \sum_{x_j \in B_l} \frac{\sigma_j}{x_k - x_j}$$

$$= \sum_{\substack{B_l \\ B_l \in \text{Near}(B_i)}} \sum_{x_j \in B_l} \frac{\sigma_j}{(x_k - x_j)} + \sum_{\substack{B_l \\ B_l \in \text{Far}(B_i)}} \sum_{x_j \in B_l} \frac{\sigma_j}{x_k - x_j}$$

$$= \sum_{\substack{B_l \\ B_l \in \text{Near}(B_i)}} \sum_{x_j \in B_l} \frac{\sigma_j}{(x_k - x_j)} + \sum_{\substack{B_l \\ B_l \in \text{Far}(B_i)}} \sum_{m=0}^p \frac{\alpha_{m, B_l}}{(x_k - c_{B_l})^m}$$

Use truncated multipole expansion of B_l instead

1 2 ... M

1	2	...	M
1	2	...	M
1	2	...	M
1	2	...	M
1	2	...	M
1	2	...	M
1	2	...	M
1	2	...	M

B_i (points to row 4, column 3)

$$\phi(x_k) = \sum_{\substack{j=1 \\ j \neq k}}^N \frac{\sigma_j}{x_k - x_j} = \sum_{\substack{B_l \in N(B_i) \\ B_l \neq B_i}} \sum_{x_j \in B_l} \frac{\sigma_j}{(x_k - x_j)} + \sum_{\substack{B_l \\ B_l \in F(B_i)}} \sum_{m=0}^p \frac{\alpha_{m, B_l}}{(x_k - c_{B_l})^m}$$

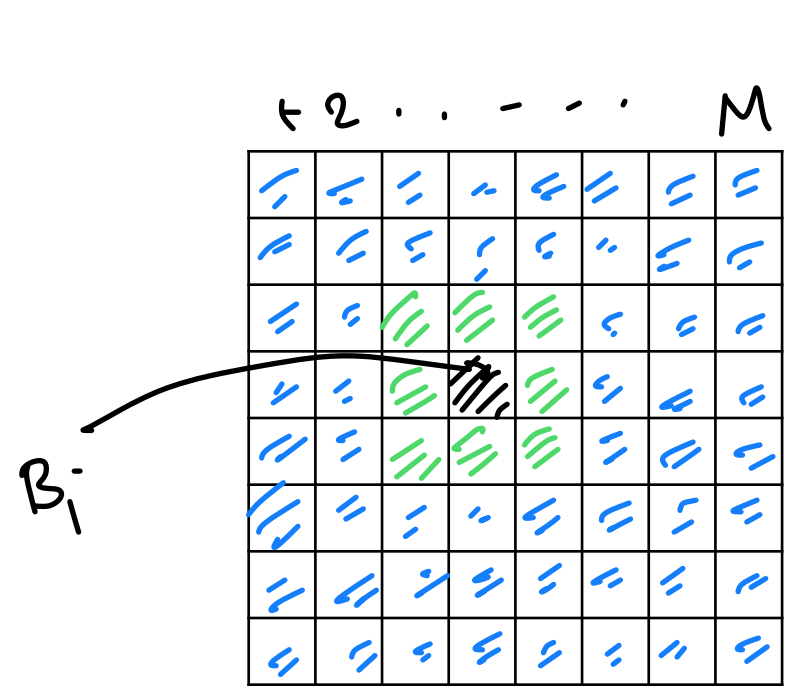
$\equiv B_j \in \text{Near}(B_i)$

$\equiv B_j \in \text{Far}(B_i)$

Algorithm

1. Compute multipole expansion coefficients (form mp)

$$\alpha_{m, B_l} = \sum_{x_j \in B_l} \sigma_j (x_j - c_{B_l})^m \text{ for each box}$$
Cost $O(Np)$
2. For $x_k \in B_i$, evaluate multipole expansions $O(N(M^2 - 9)p)$
of all boxes $B_j \in \text{Far}(B_i)$ (mpeval)
3. For $x_k \in B_i$, evaluate contribution of $O\left(N \cdot 9 \cdot \left(\frac{N}{M^2}\right)\right)$
boxes $B_j \in \text{Near}(B_i)$ directly (direct eval)



$$\phi(x_k) = \sum_{\substack{j=1 \\ j \neq k}}^N \frac{\sigma_j}{x_k - x_j} = \sum_{\substack{B_l \in N(B_i) \\ B_l \neq B_i}} \sum_{x_j \in B_l} \frac{\sigma_j}{(x_k - x_j)} + \sum_{B_l \in F(B_i)} \sum_{m=0}^p \frac{\alpha_{m, B_l}}{(x_k - c_{B_l})^m}$$

$$\equiv B_j \in \text{Near}(B_i)$$

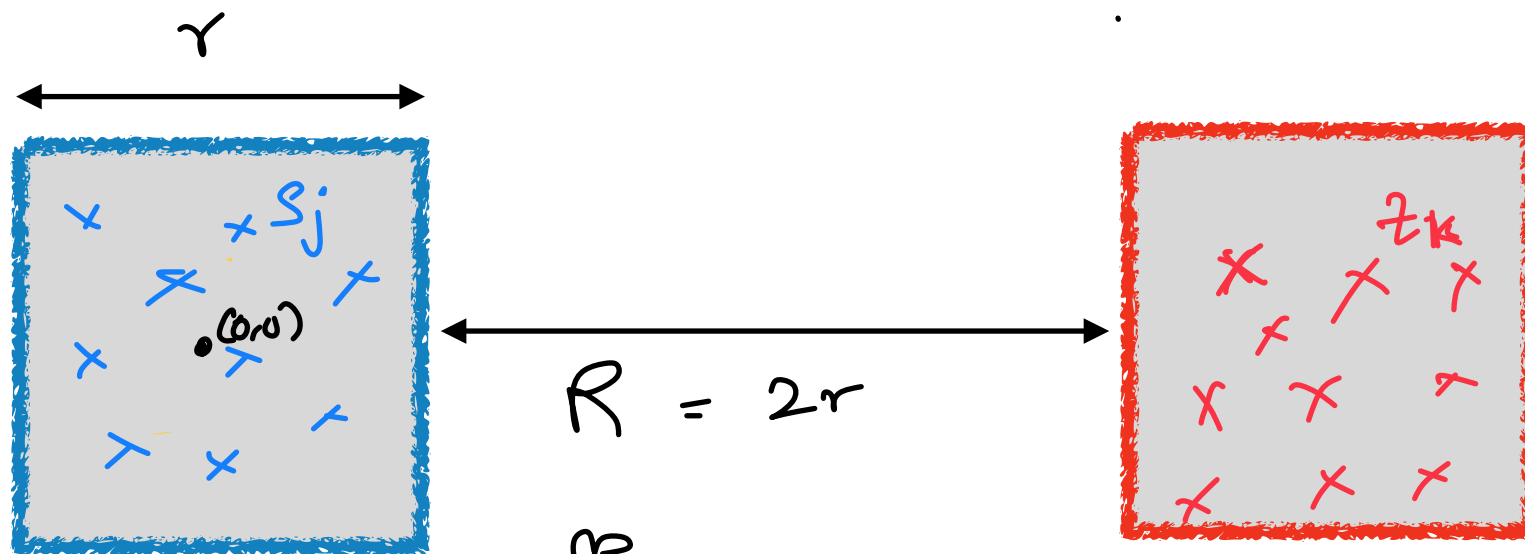
$$\equiv B_j \in \text{Far}(B_i)$$

Form mp $O(Np)$
 mpeval $O(NM^2p)$
 Direct eval $O\left(\frac{N^2}{m^2}\right)$

How to select M?
 set $NM^2p = \frac{N^2}{m^2}$
 $\Rightarrow M^4 = N$
 $\Rightarrow M^2 = N^{1/2}$

mp eval $O(N \cdot N^{1/2} p) = O(N^{1.5})$
 Direct eval $O\left(\frac{N^2}{N^{1/2}}\right) = O(N^{1.5})$
 better than $O(N^2)$!

Recall:



$$\phi(t) = \sum_{j=1}^N \frac{\sigma_j}{t - s_j} = \sum_{l=0}^{\infty} \frac{\alpha_l}{t^{l+1}}$$

Approximate ϕ by $\tilde{\phi}(t)$

$$\tilde{\phi}(t) = \sum_{l=0}^p \frac{\alpha_l}{t^{l+1}}$$

$$\text{Then } E = |\phi - \tilde{\phi}| \leq \left(\frac{\sqrt{2}}{3}\right)^{p+1} \left(\frac{1}{3r}\right)$$

$$\alpha_l = \sum_{j=1}^N \sigma_j (s_j)^l$$

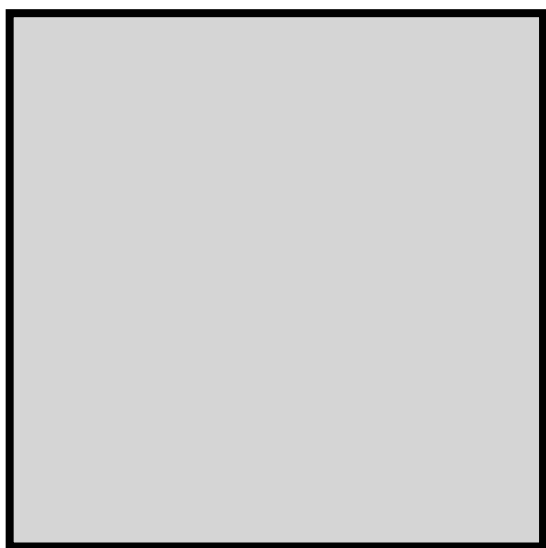
$$|t| \geq R + r = 3r$$

$$\frac{1}{|t|} \leq \frac{1}{3r}$$

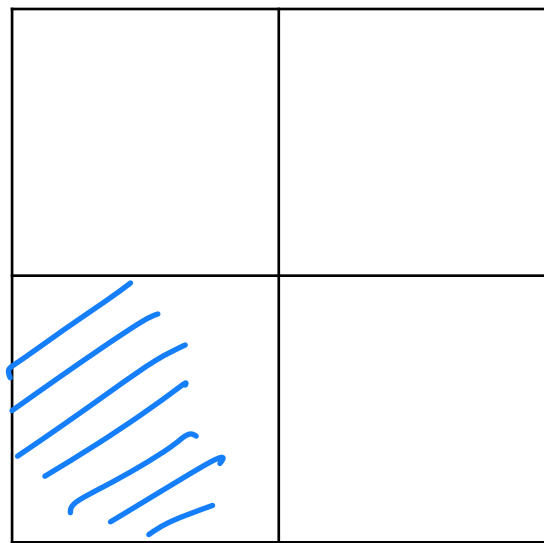
$$|s_j| \leq \sqrt{2}r$$

Error depends weakly on r^* (can be absorbed in relative error)

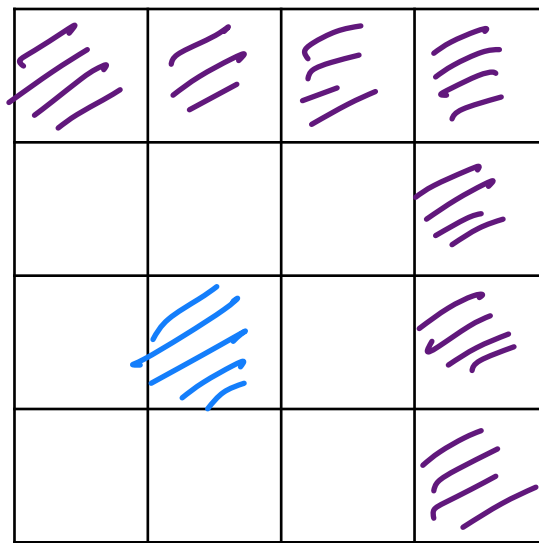
Idea: Use multipole expansions on hierarchy of scales



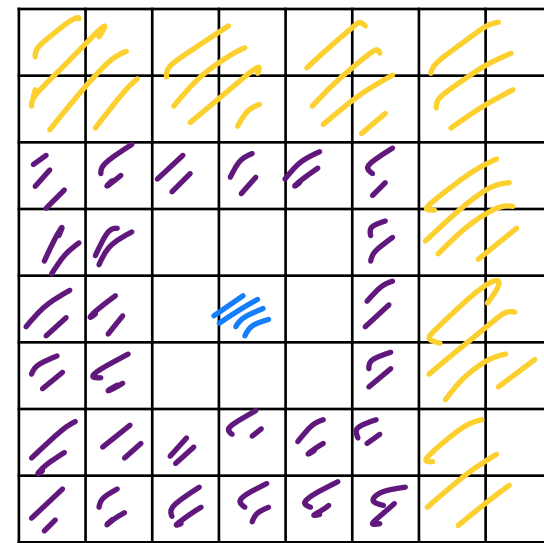
Level 0



Level 1



Level 2



Level 3 . .

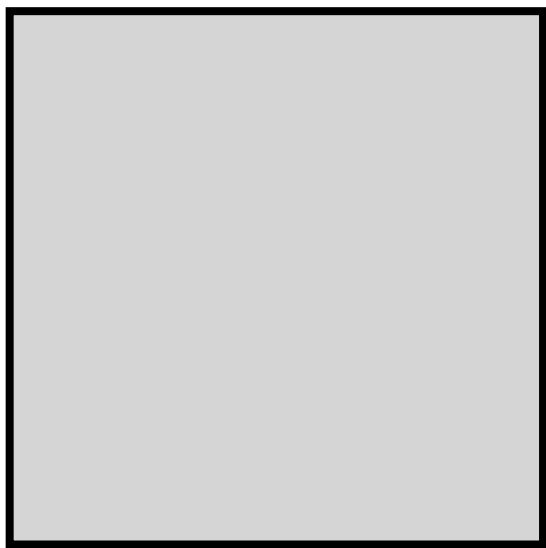
Pruned uniform quad tree: Subdivide until there are at most 's' points per box

Most boxes have already been accounted for at parent / grand parent / great grand parent .. level

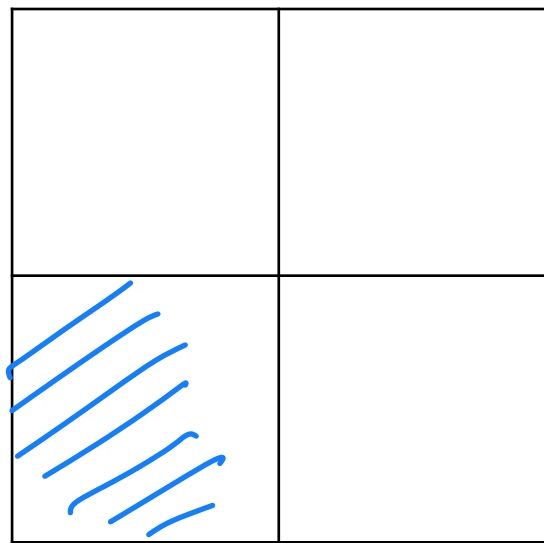
At each level only need to include contributions for $B_j \in \text{Far}(B_i)$ but $P(B_j) \notin \text{Far}(P(B_i))$

ie: $P(B_j) \in \text{Near}(P(B_i))$

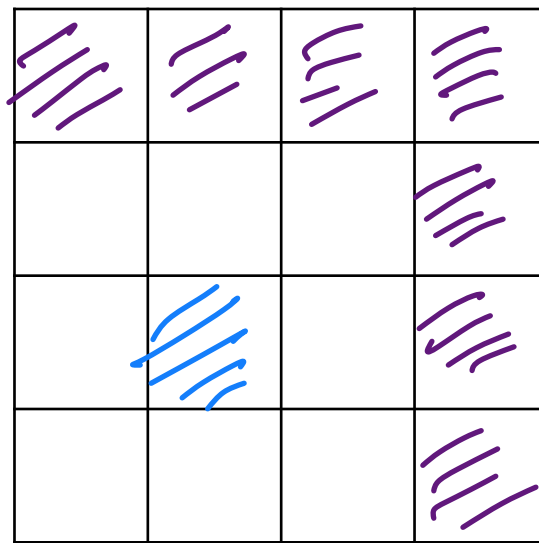
Definition: Interaction list / V-list / List 2
List of boxes at the same level which are well-separated but whose parents are neighbors
Q: How many max boxes in V-list?



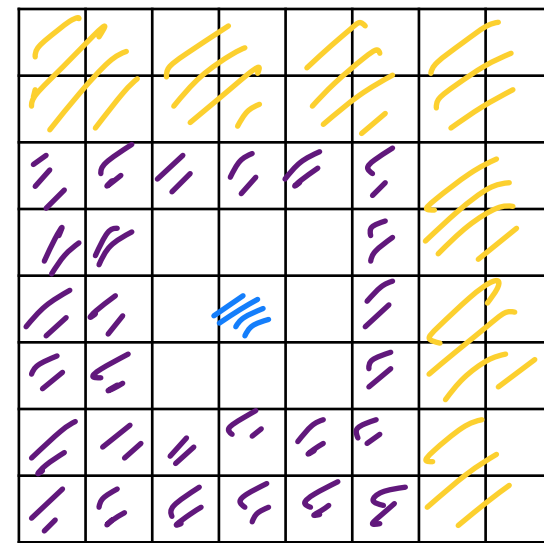
Level 0



Level 1



Level 2



Level 3 . . .

Algorithm

1. Loop over all levels ≥ 2
Form multipole expansion due to sources in that box

2. Loop over levels $L, L-1, \dots, 2$
Loop over boxes B_i at that level
Loop over targets $t_n \in B_i$
Loop over boxes $B_j \in \text{List2}(B_i)$

$$\phi(t_n) = \phi(t_n) + \sum_{B_j \in \text{List2}(B_i)} \sum_{m=0}^p \frac{\alpha_m}{(t_n - c_j)^{m+1}}$$

3. At the finest eval

$$\phi(t_n) = \phi(t_n) + \sum_{B_j \in N(B_i)} \frac{\sigma_j}{t_n - s_j}$$

$$s = p$$

Then total cost $O(N \log N p)$

cost

$$O(N \cdot L \cdot p)$$

$$O(N \cdot L \cdot 27 \cdot p)$$

$$O(9Ns)$$

$$L = \log_2 N$$

How well does it do on a computer?

2D cost $O(27 N \log N p)$

3D cost $O(189 N \log N p^2)$

6-digit accuracy

$$p = 20$$

say $\log N = 20$

Then 2D $\approx 10800 N$

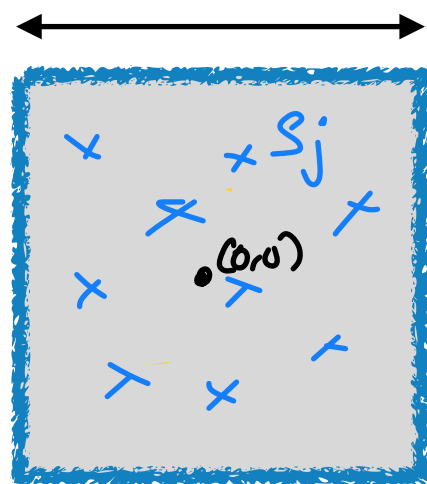
modern machines 10^{10} flops/sec
factor of 10 say for
complicated operations like
power/division
Not bad

Then speed 10^5 pts/sec

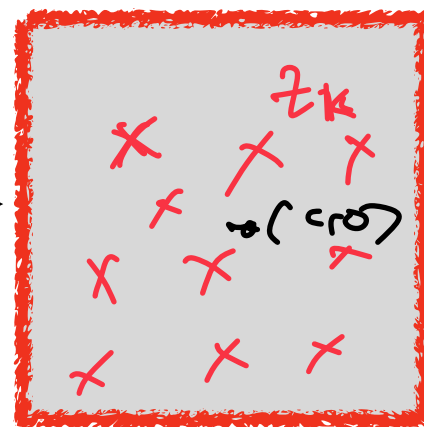
But 3D cost $= 1.6 \times 10^6 N$

Speed ≈ 1000 pts/sec/core

Back to our favorite diagram.



$$R = 2r$$



Local expansions
instead of
multipole
expansions

$$\begin{aligned} \phi(z) &= \sum_{j=1}^N \frac{\sigma_j}{z - s_j} = \sum_{l=0}^{\infty} \frac{\alpha_l}{z^{l+1}} = \sum_{l=0}^{\infty} \beta_l (z-c)^l \quad \alpha_l = \sum_{j=1}^N \sigma_j (s_j)^l \\ &= \sum_{j=1}^N \frac{\sigma_j}{z-c-(s_j-c)} = - \sum_{j=1}^N \frac{\sigma_j}{s_j-c} \frac{1}{1 - \frac{z-c}{s_j-c}} = \sum \sigma_j \frac{\sum_{l=0}^{\infty} \frac{(z-c)^l}{(s_j-c)^{l+1}}}{(s_j-c)^{l+1}} \end{aligned}$$

$$\beta_l = - \sum_{j=1}^N \frac{\sigma_j}{(s_j-c)^{l+1}} \quad |s_j-c| \geq 3r \quad |z-c| \leq \sqrt{2}r$$

$$\begin{aligned} \text{Then } E = |\phi - \tilde{\phi}| &= \left| \sum_{l=p+1}^{\infty} \beta_l (z-c)^l \right| = \left| \sum_{l=p+1}^{\infty} \sum_{j=1}^N \frac{\sigma_j (z-c)^l}{(s_j-c)^{l+1}} \right| \\ &\leq \sum_{j=1}^N \frac{|\sigma_j|}{3r} \sum_{l=p+1}^{\infty} \left(\frac{|z-c|}{|s_j-c|} \right)^l \leq \sum_{j=1}^N \frac{|\sigma_j|}{3r} \sum_{l=p+1}^{\infty} \left(\frac{\sqrt{2}r}{3r} \right)^l \\ &\leq \frac{\sum |\sigma_j|}{3r} \left(\frac{\sqrt{2}}{3} \right)^{p+1} \left(\frac{1}{1 - \frac{\sqrt{2}}{3}} \right) \end{aligned}$$

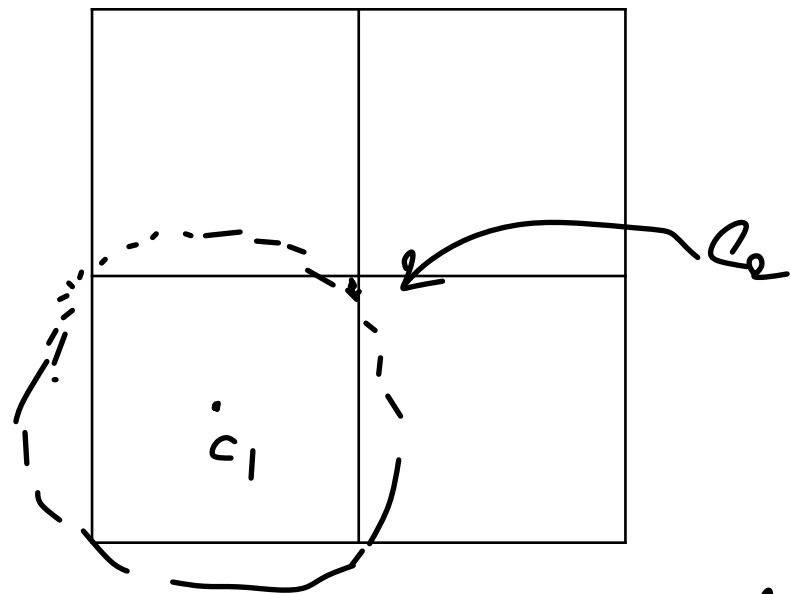
Using only local expansion instead of multipole expansions won't change anything

FMM - Key insights

Avoid forming expansions at every level & use translation operators

$$\phi(z) = \sum_{s \in B_1} \frac{\sigma_j}{z - s_j} = \sum_{l=0}^{\infty} \frac{\alpha_l}{(z - c_1)^{l+1}}$$

$$\alpha_l = \sum_{s_j \in B_1} \sigma_j (s_j - c)^l$$



$$\frac{1}{(z - z_0)^k} = \sum_{l=k}^{\infty} \binom{l-1}{k-1} \frac{z_0^{l-k}}{z^l}$$

$$\sum_{l=0}^{\infty} \frac{\alpha_l}{(z - c_1)^{l+1}} = \sum_{l=0}^{\infty} \frac{\alpha_l}{(z - c_2 - (c_1 - c_2))^{l+1}}$$

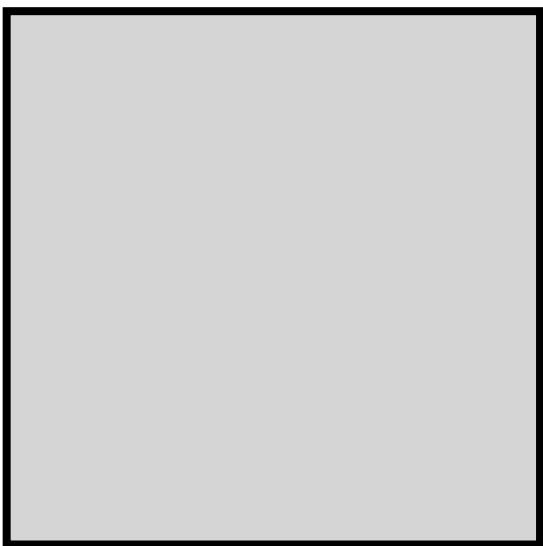
$$= \sum_{l=0}^{\infty} \alpha_l \sum_{m=l}^{\infty} \binom{m-1}{l-1} \frac{(c_1 - c_2)^{m-l}}{(z - c_2)^m}$$

$$= \sum_{m=0}^{\infty} \frac{1}{(z - c_2)^m} \sum_{l=0}^m \binom{m-1}{l-1} \alpha_l (c_1 - c_2)^{m-l}$$

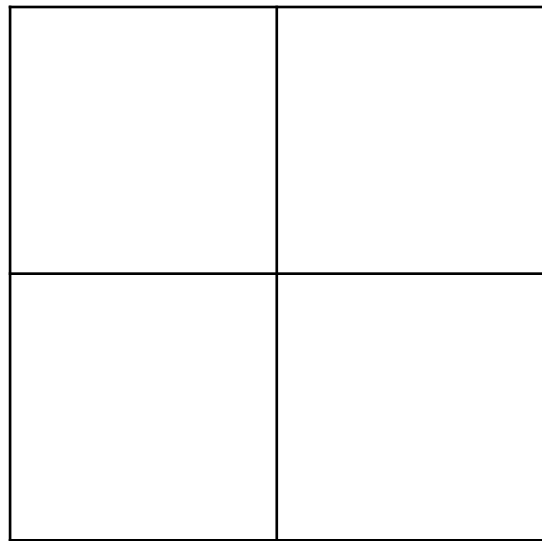
i.e. $\mathcal{I}_B \phi(z) = \sum_{m=0}^{\infty} \frac{\beta_m}{(z - c_2)^{m+1}}$
 Then $\beta_m = \sum_{l=0}^m \binom{m-1}{l-1} \alpha_l (c_1 - c_2)^{m-l}$
 m2m operator

Similar translation operators for m2L & L2L

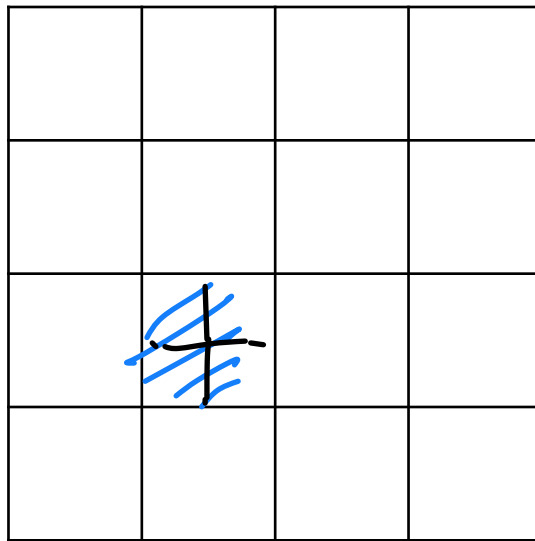
multipole expansion coefficients of parent box



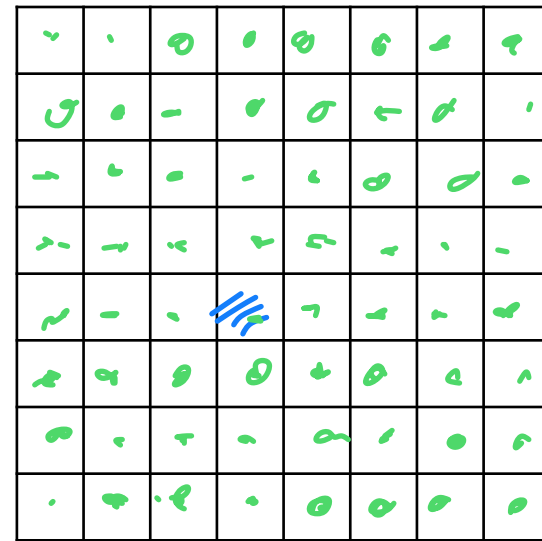
Level 0



Level 1

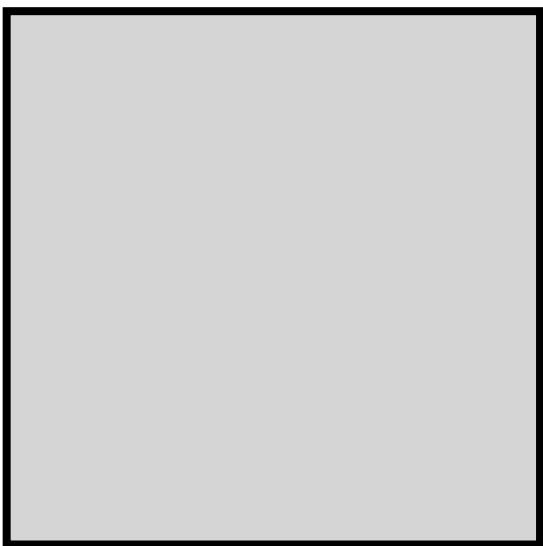


Level 2

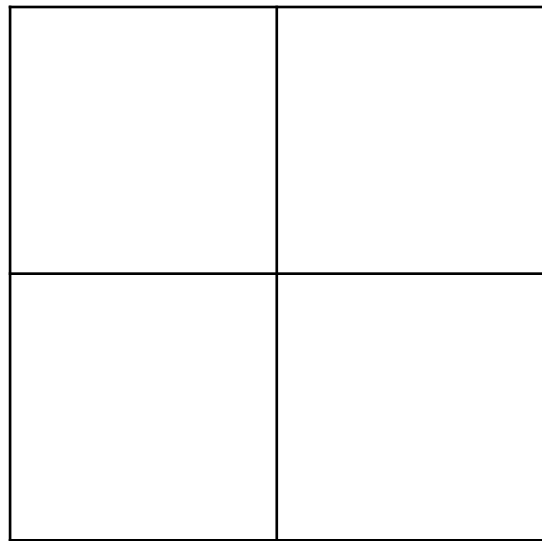


Level 3

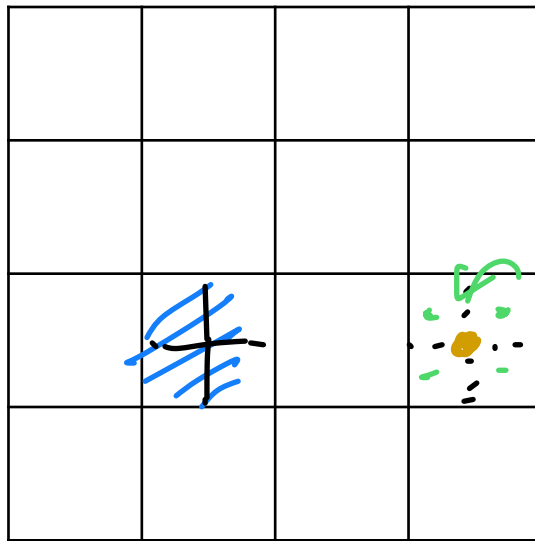
1. Form multipole expansion for each leaf box $O(N_p)$



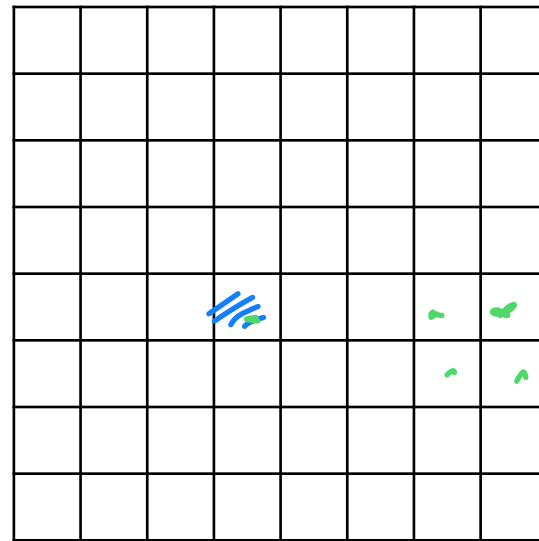
Level 0



Level 1



Level 2

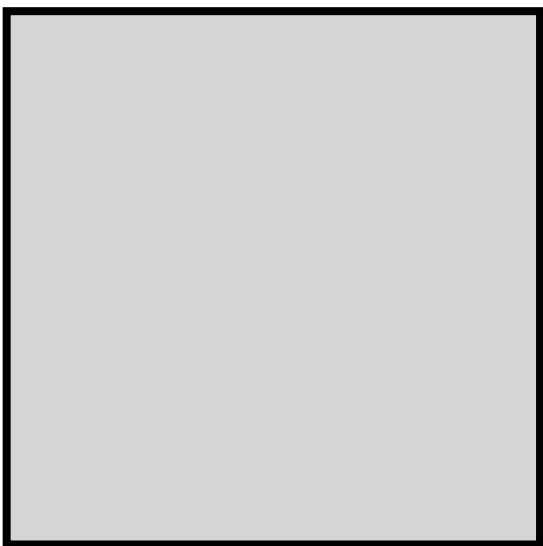


Level 3

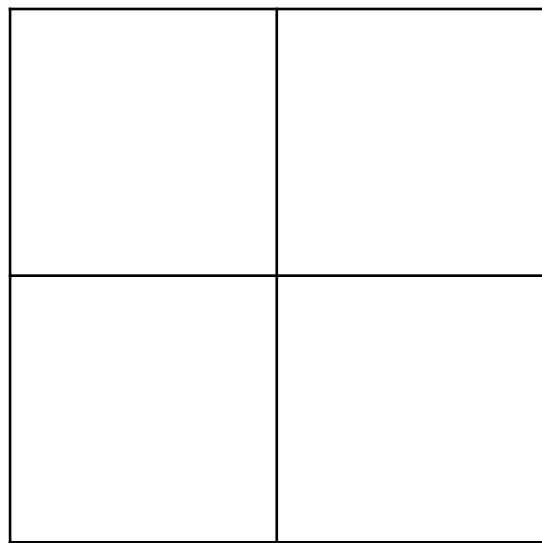
1. Form multipole expansion for each leaf box
2. Merge multipole expansions of children boxes to form multipole expansions of parents

$$O(Np)$$

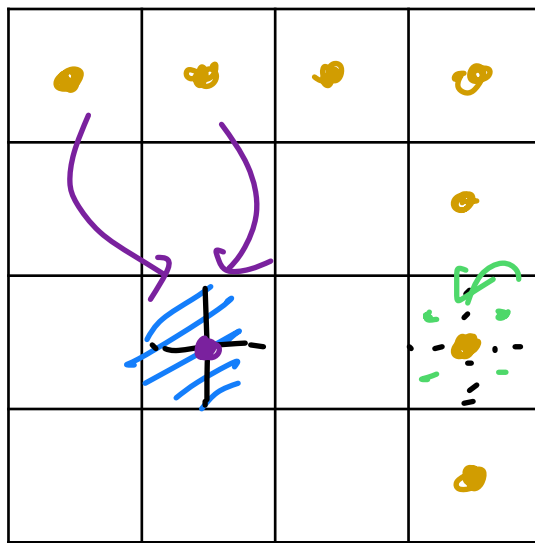
$$O(N_B p^2)$$



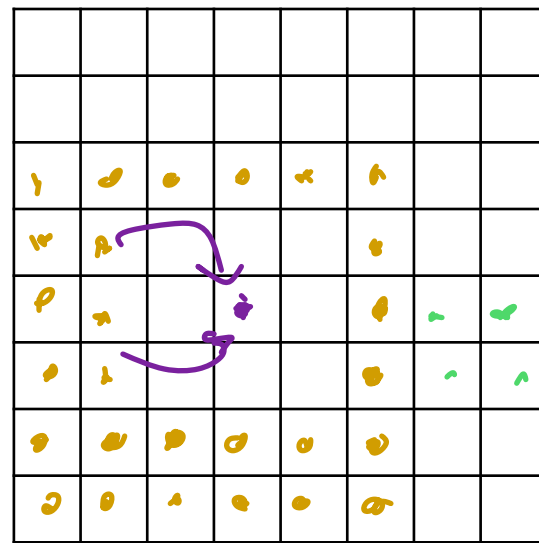
Level 0



Level 1



Level 2



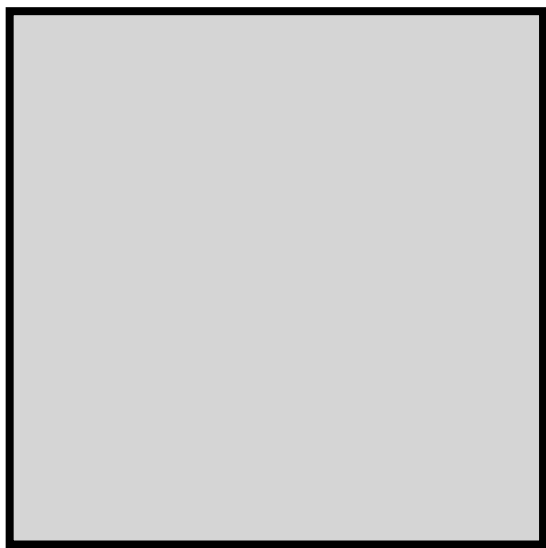
Level 3

1. Form multipole expansion for each leaf box
2. Merge multipole expansions of children boxes to form multipole expansions of parents
3. Convert multipole expansion from all boxes in interaction list to a local expansion

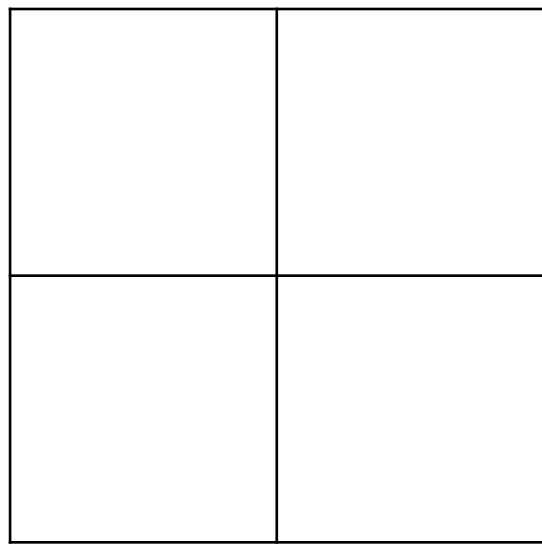
$$O(Np)$$

$$O(N_B p^2)$$

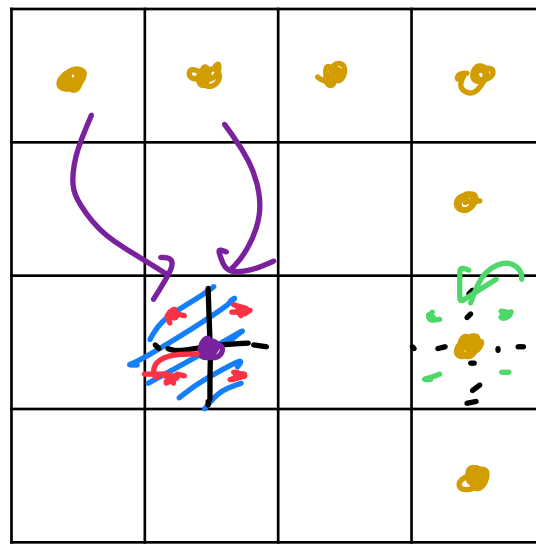
$$O(27 N_B p^2)$$



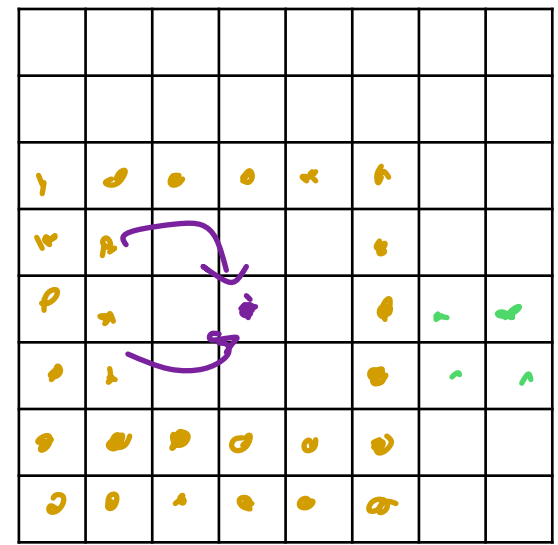
Level 0



Level 1



Level 2



Level 3

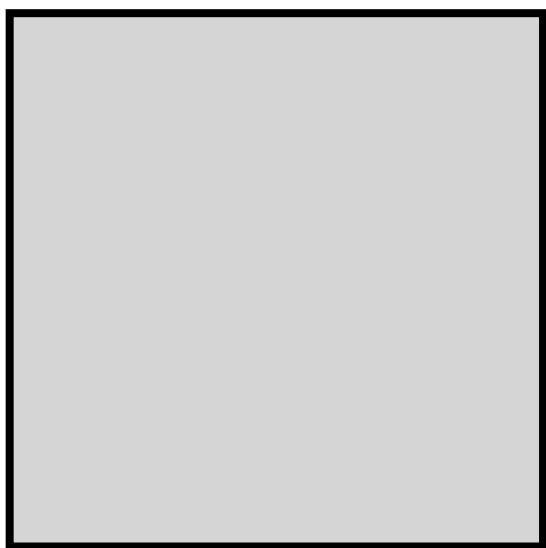
1. Form multipole expansion for each leaf box
2. Merge multipole expansions of children boxes to form multipole expansions of parents
3. Convert multipole expansion from all boxes in interaction list to a local expansion
4. Convert local expansion of parent box to local expansion of children boxes

$$O(Np)$$

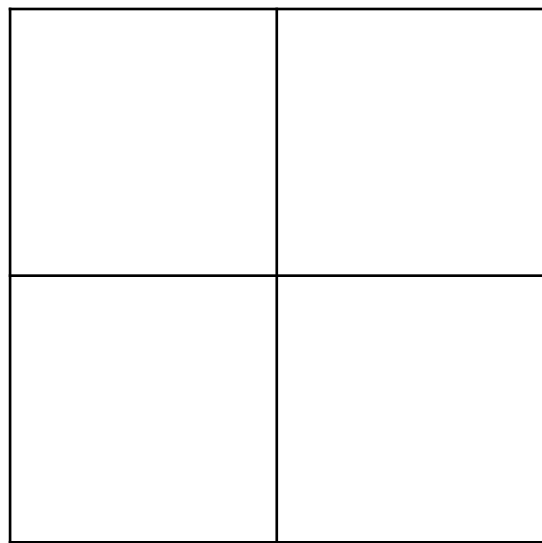
$$O(N_B p^2)$$

$$O(27 N_B p^2)$$

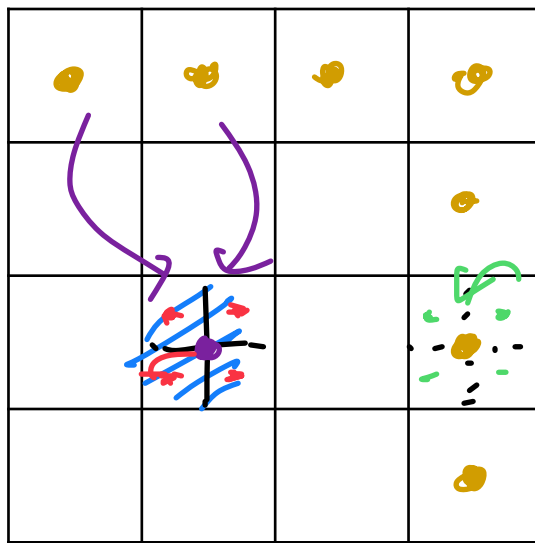
$$O(N_B p^2)$$



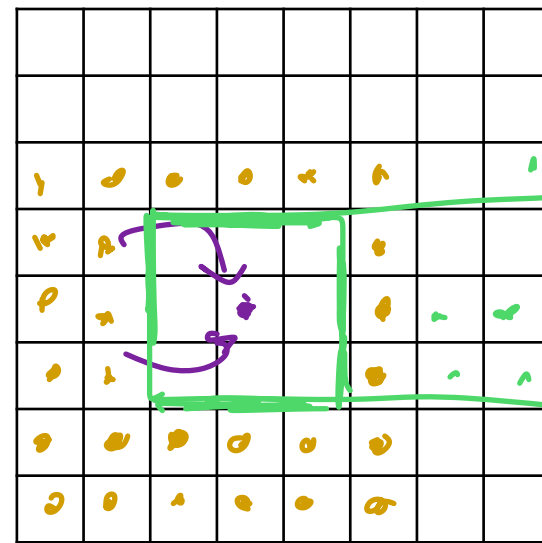
Level 0



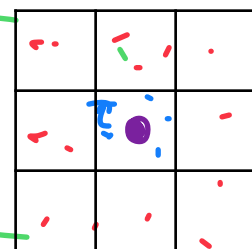
Level 1



Level 2



Level 3



1. Form multipole expansion for each leaf box

$$O(N_p)$$

2. Merge multipole expansions of children boxes to form multipole expansions of parents

$$O(N_B p^2)$$

3. Convert multipole expansion from all boxes in interaction list to a local expansion

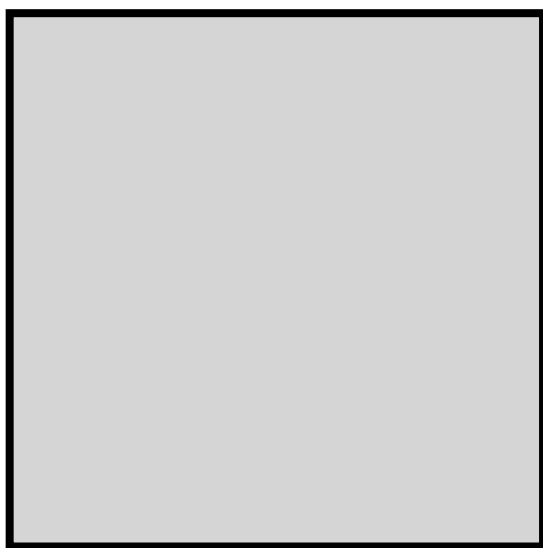
$$O(27 N_B p^2)$$

4. Convert local expansion of parent box to local expansion of children boxes

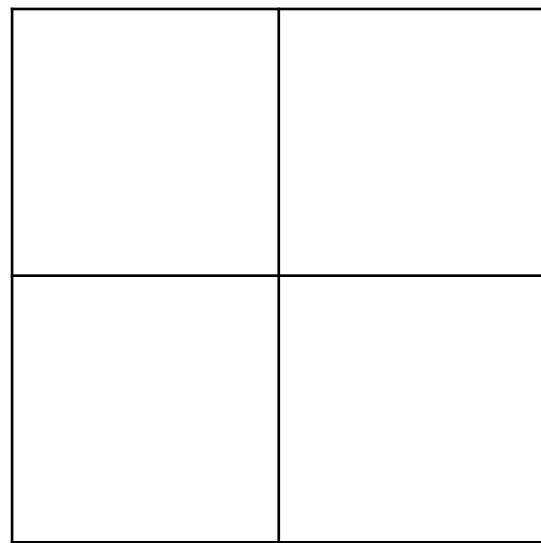
$$O(N_B p^2)$$

5. Evaluate local expansion in leaf box at target points

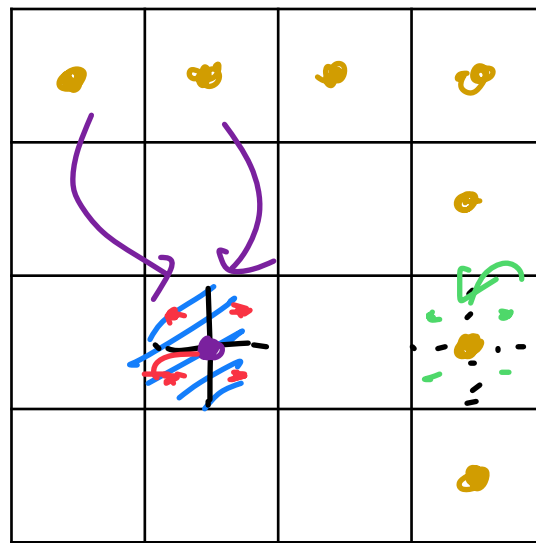
$$O(N_p)$$



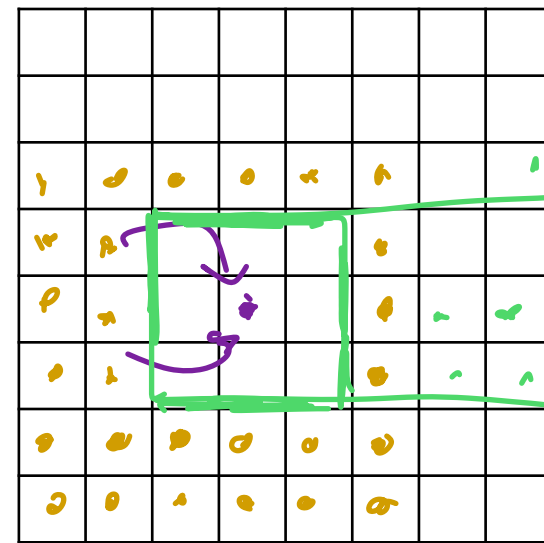
Level 0



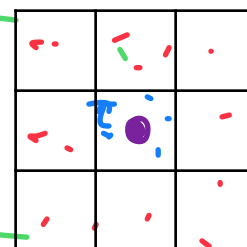
Level 1



Level 2



Level 3



1. Form multipole expansion for each leaf box

$$O(Np)$$

2. Merge multipole expansions of children boxes to form multipole expansions of parents

$$O(N_B p^2)$$

3. Convert multipole expansion from all boxes in interaction list to a local expansion

$$O(27 N_B p^2)$$

4. Convert local expansion of parent box to local expansion of children boxes

$$O(N_B p^2)$$

5. Evaluate local expansion in leaf box at target points

$$O(Np)$$

6. Include contributions from neighbor boxes directly

$$9 N_B$$

$$N_B = N/8 \quad \text{Choose } 8 = p$$

Barnes Hut

Formmp	$N \log N p$
mpeval	$27 N \log N p$
direct	$9 N p$

Total $28 N \log N p + 9 N p$

3D:
 $190 N \log N p^2 + 27 N p^2$

est speed 1000 pts/sec

FM M

Formmp	$- N p$
mpmp	$- N p$
mp $\rightarrow$ loc	$- 27 N p$
loc $\rightarrow$ loc	$- N p$
loc-eval	$- N p$
direct	$- 9 N p$
Total	$- 40 N p$

almost
factor of
log N better

$218 N p^{3/2} + 2 N p^2$

33000 pts/sec